



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
13-12-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The heat absorbed or rejected during a polytropic process is ($PV^n = C$) $\left(\gamma = \frac{C_p}{C_v}\right)$

(A) $\left(\frac{\gamma - n}{\gamma - 1}\right) \times \text{work done}$

(B) $\left(\frac{\gamma - n}{\gamma - 1}\right)^2 \times \text{work done}$

(C) $\left(\frac{\gamma - n}{\gamma - 1}\right)^{1/2} \times \text{work done}$

(D) $\left(\frac{\gamma - n}{\gamma - 1}\right)^3 \times \text{work done}$

2. The dynamic viscosity of an oil, used for lubrication between a shaft and sleeve is 6 poise. The shaft is of diameter 0.4 m and rotates at 190 r.p.m. Calculate the power lost in the bearing for a sleeve length of 90 mm. The thickness of the oil film is 1.5 mm.

(A) 516.44 W

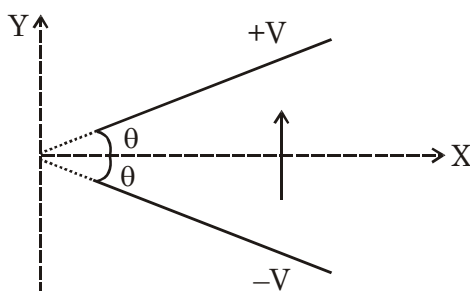
(B) 358.24 W

(C) 716.48 W

(D) 658.48 W

Space for Rough Work

3. An electric dipole $\vec{P}$, pointing in the y-direction, is placed midway between two large conducting plates as shown in figure. Each plate makes a small angle θ with respect to x-axis, and they are maintained at potential $\pm V$. The direction of net force on dipole will be :

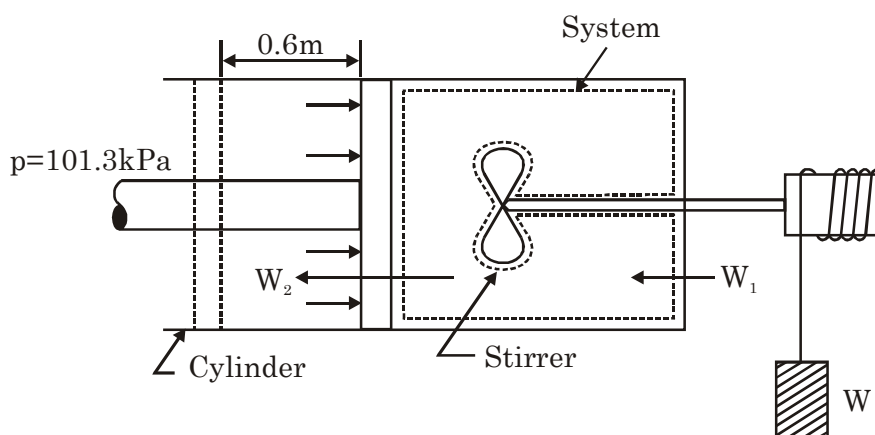


- (A) Along +X-axis (B) Along -ve X-axis (C) Zero (D) Along -ve Y-axis
4. Assume an imaginary planet made up of a material which directly evaporates from solid state (sublimate). The energy of sublimation is assumed to be provided by the reduction in gravitational self energy of the planet. If heat of sublimation is H J/kg. What must be the minimum radius of planet which will result in self sublimation of planet. [ρ = density of planet]

(A) $R_{\min} = \sqrt{\frac{4}{3} \frac{H}{G\rho\pi}}$ (B) $R_{\min} = \sqrt{\frac{3H}{4G\rho\pi}}$ (C) $R_{\min} = \sqrt{\frac{H}{G\rho\pi}}$ (D) $R_{\min} = \frac{3}{2} \sqrt{\frac{H}{G\rho\pi}}$

Space for Rough Work

5. A dummy Earth is made for one of Spielberg's films. It has the same size and mass as the real Earth, but its construction is different: there is a small ball of extremely dense matter inside a very light outer plastic ball. Due to some inaccuracy during assembly, the center of mass of the heavy ball is shifted in the equatorial plane by a distance $d = 100$ km from the center of the outer shell. Find the minimal period of revolution of satellite orbit in the equatorial plane. Take : Radius of earth = 6400 km & time period for a satellite in near earth orbit as 5060 sec.
- (A) 5062 sec approx. (B) 5058 sec approx.
(C) 5075 sec approx. (D) 5034 sec approx.
6. A piston and cylinder machine containing a fluid system has a stirring device as shown in figure. The piston is frictionless, and it is held down against the fluid due to atmospheric pressure of 101.3 kPa. The stirring device is turned 9500 revolutions with an average torque against the fluid of 1.25 Nm. Meanwhile the piston of 0.65 m diameter moves out 0.6 m. Find the net work transfer for the system.



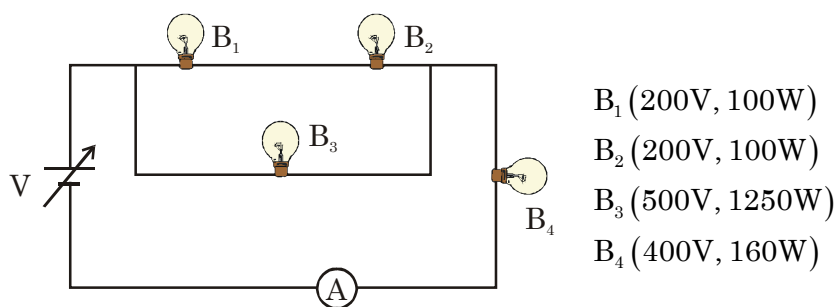
- (A) -27.227 kJ (B) -54.455 kJ (C) -74.622 kJ (D) -96.789 kJ

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

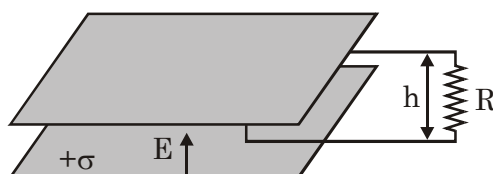
7. Figure shows a combination of 4 electric bulb joined with a d.c. source whose voltage can be varied. Choose the **CORRECT** statement(s) :



- (A) When V is slowly increased, bulb B_1 & B_2 will fuse first.
 (B) When V is slowly increased, bulb B_3 will fuse first.
 (C) When V is slowly increased, bulb B_4 will fuse first.
 (D) Maximum power output from the system will be 185.6 watt.

Space for Rough Work

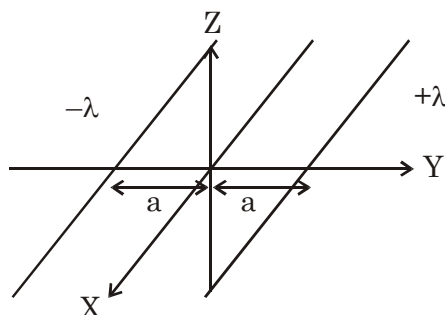
8. Assuming Sun to be a mass of 1.99×10^{30} kg uniform sphere of radius $R = 6.96 \times 10^8$ m, choose the **CORRECT** statement(s) :
- (A) The gravitational energy of Sun will be 2.28×10^{41} J.
- (B) On increasing the radius of sun, its gravitational energy will decrease if mass of the sun remains constant.
- (C) Assume sun to radiate at a rate of 3.86×10^{26} W, & if all this come from gravitational energy, the sun will last for approx. 5.90×10^{14} s.
- (D) If the density of sun remains same, then its gravitational energy will increase by 32 times if its radius is increased by a factor of 2.
9. A stationary electric dipole $\vec{p} = p\hat{k}$ is situated at the origin. A positive point charge q (mass m) executes circular motion (radius a) at constant speed in the field of the dipole. Choose the **CORRECT** statement(s) :
- (A) The situation can't be fulfilled.
- (B) The plane of orbit will be x-y plane with negative z-coordinate.
- (C) Angular momentum of the mass is $\sqrt{\frac{qpm}{3\sqrt{3}\pi\epsilon_0}}$ about z-axis.
- (D) Total energy of the charge will be equal to $\frac{qp}{6\sqrt{3}\pi\epsilon_0 a^2}$
10. A rectangular loop of wire is situated so that one end is between the plates of a parallel-plate capacitor oriented parallel to the field $\vec{E}$. The other end is way outside where the field is essentially zero.



- (A) The emf in the loop is zero.
- (B) The current through the resistance R is zero.
- (C) To explain the arrangement on energy analysis we must take fringe effect into account.
- (D) The emf in the loop is $\frac{\sigma}{\epsilon_0} h$.

Space for Rough Work

11. Two infinitely long wires running parallel to the x-axis carry uniform charge densities $+\lambda$ & $-\lambda$ as shown. Choose the **CORRECT** statement(s) :-



- (A) Potential at any point (x, y, z) is $\frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{(y+a)^2}{(y-a)^2} \right]$
- (B) Potential at any point (x, y, z) is $\frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{(y+a)^2 + z^2}{(y-a)^2 + z^2} \right]$
- (C) Equipotential surface will be spherical with radius $R = 2a \times \frac{1}{e^{\frac{\pi\epsilon_0 V_0}{\lambda}} - e^{\frac{-2\pi\epsilon_0 V_0}{\lambda}}}$ [V_0 is potential of given equipotential surface]
- (D) Equipotential surface will be cylindrical with radius as $R = \frac{a}{e^{\frac{2\pi\epsilon_0 V_0}{\lambda}} - e^{\frac{-2\pi\epsilon_0 V_0}{\lambda}}}$ [V_0 is potential of given equipotential surface]

Space for Rough Work

12. When two slabs of n-type and p-type semiconductors are put in contact, the relative affinities of the materials cause electrons to migrate out of the n-type material across the junction to the p-type material. This leaves behind a volume in the n-type material that is positively charged and creates a negatively charged volume in the p-type material. Let us model this as two infinite slabs of charge, both of thickness a with the junction lying on the plane $z = 0$. The n-type material lies in the range $0 < z < a$ and has uniform charge density $+\rho_0$. The adjacent p-type material lies in the range $-a < z < 0$ and has uniform charge density $-\rho_0$. Thus:

$$\rho(x, y, z) = \rho(z) = \begin{cases} +\rho_0 & 0 < z < a \\ -\rho_0 & -a < z < 0 \\ 0 & |z| > a \end{cases}$$

- (A) The electric field for $-a < z < a$ is $\frac{-\rho_0(a+z)}{\epsilon_0} \hat{k}$
- (B) The electric field for $0 < z < a$ is $\frac{-\rho_0(a+z)}{\epsilon_0} \hat{k}$
- (C) Potential difference between two points P_1 (located at $z > a$), P_2 (located at $z < -a$) will be positive.
- (D) Electric field in the region $|z| > a$ will be zero.

Space for Rough Work

SECTION-II : (Maximum Marks : 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

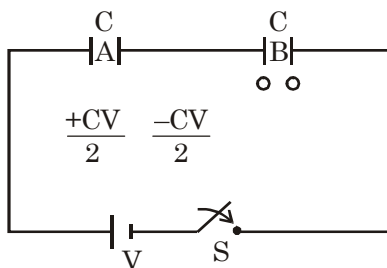
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+										-									
●	●	0	0	0	0	0	0	0	0	●	●	●	0	0	●	●	●	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

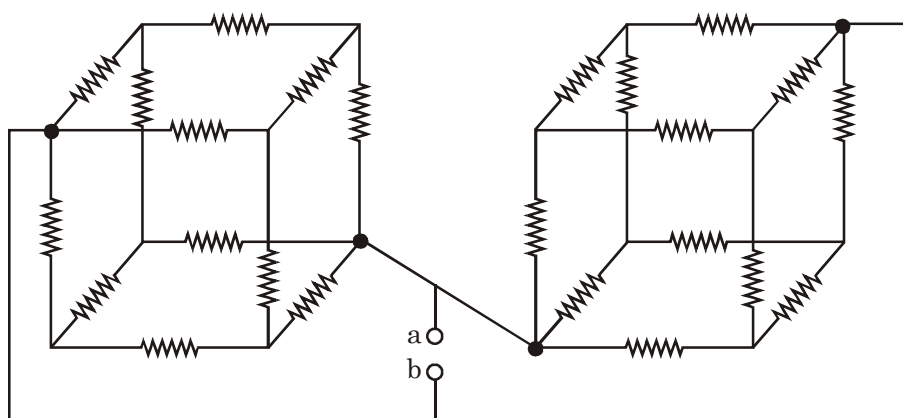
1. Initially S is open and capacitor A is charged up to potential difference $\frac{V}{2}$ and capacitor B is

uncharged. After closing the switch S the heat loss in the circuit is $\frac{CV^2}{2 \times n}$ J. Find n?



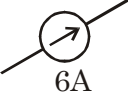
Space for Rough Work

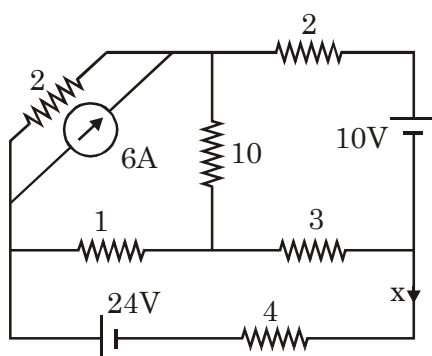
2. Find the equivalent resistance (in Ω) between terminals a and b of the circuit shown in figure. Each resistor has a value of 1Ω .



3. A vertical gap 2.2 cm wide of infinite extent contains a fluid of viscosity 2.0 N s/m^2 and specific gravity 0.9. A metallic plate $1.2 \text{ m} \times 1.2 \text{ m} \times 0.2 \text{ cm}$ is to be lifted up with a constant velocity of 0.15 m/sec , through the gap. If the plate is in middle of the gap, find the force required. The weight of the plane is 40 N .

Space for Rough Work

4. The conductors of a two-wire transmission line (4 km long) are spaced 45 cm between centre. If each has a diameter of 1.5 cm, the capacitance of the line is α nF. Fill the value of α .
(Given : $\ln 3 = 1.0986$, $\ln 5 = 1.6094$, $\ln 2 = 0.6931$)
5. Consider two spheres of radius a , immersed deep in the sea and held quite far apart. If the potential difference between them is V , and current measured to be I , then conductivity of sea water is given by σ S.I. unit. Find the value of σ . [Given : $I = 3$ A, $V = 10$ Volt & $a = 1$ cm]
6. Determine the current x (in A) in the 4Ω resistance of the circuit shown in figure. The  represents constant current carrying source of 6A.



Space for Rough Work

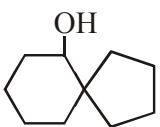
PART-2 : CHEMISTRY

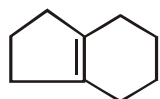
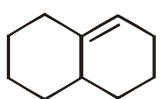
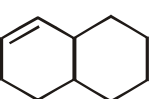
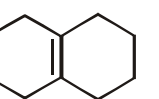
SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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1. Which of the following structure is non-planar.

- (A) Inorganic graphite sheet (B) $\text{Si}_3\text{O}_9^{6-}$
 (C) $\text{B}_3\text{O}_6^{3-}$ (D) I_2Cl_6

2.  $\xrightarrow[\Delta]{\text{H}_2\text{SO}_4}$ A Major product A is :
 (Major)

- (A)  (B)  (C)  (D) 

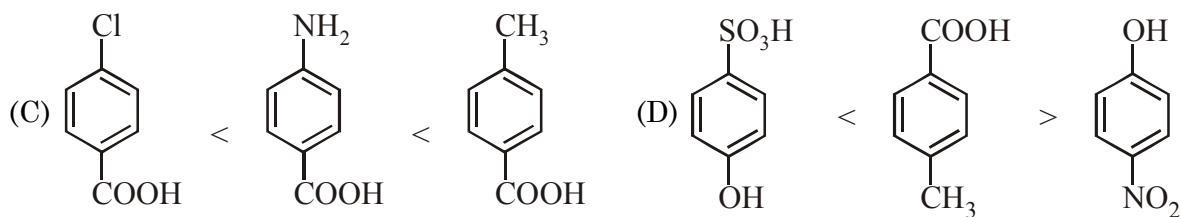
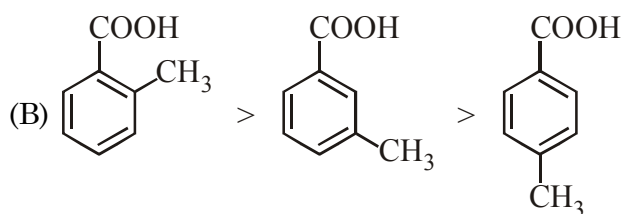
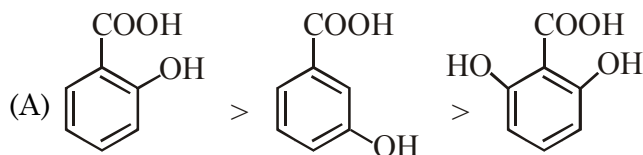
3. A certain non-volatile electrolyte contains 40% carbon, 6.7% hydrogen and 53.3% oxygen. An aq. solution containing 5.7% by mass of solute boils at 100.17°C . Determine the molecular formula of the compound ($K_b = 0.51^\circ\text{C/m}$).

- (A) CH_3OH (B) HCHO (C) $\text{C}_2\text{H}_5\text{OH}$ (D) $\text{C}_6\text{H}_{12}\text{O}_6$

Space for Rough Work

4. In protonation of NH_3 change occurs in :
- (A) Hybridisation of N-atom and electronic geometry
 (B) Shape of molecule and H-N-H bond angle
 (C) Neither in hybridisation nor in shape of molecule
 (D) Number of hybrid orbitals of N and total valence electrons

5. Correct order of acidic strength among the following :



6. An atomic solid crystallizes in a body centre cubic lattice and the surface of the atoms at the adjacent corner are separated by 67.0 pm. If the atomic mass of A is 75, then density of the solid is nearly - ($N_A = 6 \times 10^{23}$)

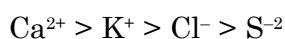
- (A) 2.00 gm/cm^3 (B) 1.75 gm/cm^3 (C) 2.5 gm/cm^3 (D) 4.0 gm/cm^3

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

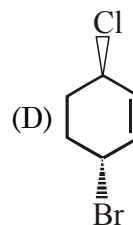
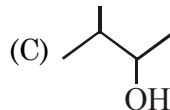
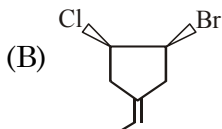
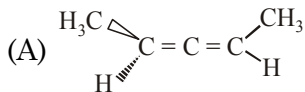
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7. Consider the order :



Select **INCORRECT** statement(s).

- (A) Decreasing order of number of electrons in $n = 3$
- (B) Decreasing order of size
- (C) Decreasing order of IE
- (D) Decreasing order of Z_{eff}
8. Which of the following compounds are optically active



Space for Rough Work

9. In the following gaseous phase first order reaction :
 $A(g) \rightarrow 2B(g) + C(g)$
Initial pressure was found to be 400 mm of Hg and it changed to 1000 mm of Hg after 20 min. then -
(A) Half life for A is 10 min.
(B) Rate constant is 0.0693 min^{-1}
(C) Partial pressure of C at 30 min is 350 mm of Hg
(D) Total pressure after 30 min is 1100 mm of Hg.
10. Which among the following is **CORRECT**?
(A) $I(CN)_2^-$ is linear species because I and C atoms are sp hybridised
(B) Each bond length is identical present in SF_2 , CO_3^{2-} and HCO_2^-
(C) $IO_2F_2^-$ and SOF_4 both have atleast one bond angle less than 90°
(D) The number of angles less than 120° present in PH_2F_3 is 8
11. Which of the following gives efferevsence of CO_2 when treated with $NaHCO_3$.
(A) Aspirine
(B) Benzene sulphonic acid
(C) o-cresol
(D) quinol
12. Regarding non-ideal behaviour of gases, choose the correct statement(s) :
(A) Gas which behave ideally over a large range of pressure at Boyle's temperature is easy to liquify
(B) Vanderwaal constant 'b' is larger for 'Ar' than 'Ne' gas
(C) The value of Boyle's temperature increases with the strength of attractive interaction between molecules
(D) Pressure exerted by 1 mole O_2 is greater than 1 mole H_2O at 350K if both gases are present in same volume.

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SECTION-II : (Maximum Marks : 24)

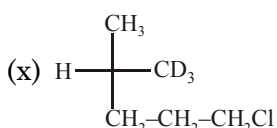
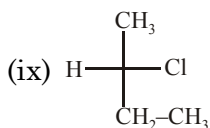
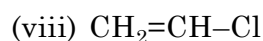
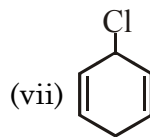
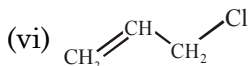
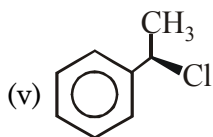
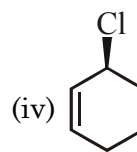
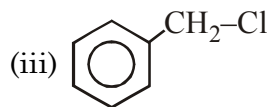
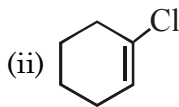
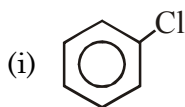
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+					-				
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

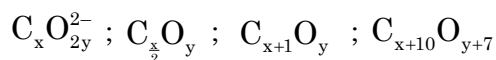
- For the given species find the summation of number of atoms present in the plane containing maximum number of atoms.
 IF_5 , IF_7 , SF_6 , CCl_4 , SO_3
- Number of compounds which slowly racemises on addition of SbCl_5 .



Space for Rough Work

3. The vapour pressure of two pure liquids A and B that from an ideal solution are 100 and 900 torr respectively at temperature T. This liquid solution of A and B is composed of 1 mole of A and 1 mole of B. What will be the pressure (in torr), when 1 mole of mixture has been vaporized.

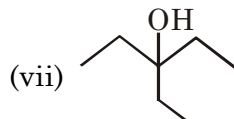
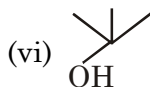
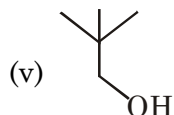
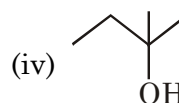
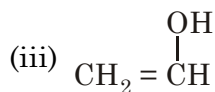
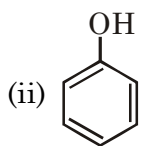
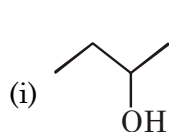
4. Consider the following four species :



If 'x = y = 2' then calculate the value of |p-q| where 'p' and 'q' are total number of sp² and sp hybridised. Carbon atom respectively present in the given four species.

Space for Rough Work

5. How many of the following will not give immediate turbidity on treatment with anhydrous ZnCl_2 and conc. HCl



6. A sample of air at 87°C and 81% relative humidity is cooled to 27°C . The mass of water vapour (in gm) condensed per 16 litre of air is (vapour pressure of water at 27°C and 87°C are 25 mm and 600 mm Hg , respectively and $R = 0.08 \text{ l-atm /K-mol}$).

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 18)**

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

-
1. If eccentricity of conjugate hyperbola of given hyperbola $\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2} = 3$ is e' , then the value of $8e'$ is
- (A) 5 (B) 8 (C) 10 (D) 12
2. Let a circle with centre O and radius k. Three points P, Q, R are chosen on a circle such that if length of chord PQ, QR, RP are x, y & z respectively, $x^2 + y^2 + z^2 = 5$, $\angle QOR = 60^\circ$ and Area of ΔPQR is $\frac{\sqrt{3}}{4}$ unit, then the value of radius k of circle is equal to
- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

3. Let $\frac{x}{2}\cos\theta + \frac{y}{1}\sin\theta = 1$, where $\theta = \frac{\pi}{2020}$ intersects the ellipse $x^2 + 2y^2 = 6$ at P and Q, then angle between tangents at P and Q to the ellipse is -
- (A) $\frac{\pi}{16}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$
4. In a triangle ABC, the equation of the angle bisector of angle B is $y = x$. If $A \equiv (2, 6)$ and $B \equiv (1, 1)$. The length of perpendicular distance of BC from the origin is equal to
- (A) 4 (B) $\frac{4}{\sqrt{21}}$ (C) $\frac{4}{\sqrt{26}}$ (D) 5
5. The value of $\cos\frac{2\pi}{2019} \cdot \cos\frac{4\pi}{2019} \cdot \cos\frac{6\pi}{2019} \dots \cos\frac{2018\pi}{2019}$ is
- (A) 2^{-1010} (B) 2^{-1009} (C) 2^{-1011} (D) 2^{-2018}
6. In the expansion of $(1 + 5x + 6x^2)^{10} = a_0 + a_1x + \dots + a_{20}x^{20}$, the numerically greatest coefficient is-
- (A) ${}^{10}C_7 \cdot {}^{10}C_8 \cdot 2^3 \cdot 3^2$ (B) ${}^{10}C_7 \cdot {}^{10}C_8 \cdot 2^7 \cdot 3^8$ (C) ${}^{10}C_8 \cdot {}^{10}C_9 \cdot 2^8 \cdot 3^9$ (D) ${}^{10}C_5 \cdot {}^{10}C_5 \cdot 2^5 \cdot 3^5$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Let $f : \mathbb{R} \rightarrow \left[-\frac{\pi}{4}, \frac{\pi}{2} \right)$ be a function defined by $f(x) = \tan^{-1} \left(\tan^{-1}(\alpha) + x^4 - \frac{7}{4} - x^2 \right)$.

If f is surjective, then -

(A) $\alpha + \frac{1}{\alpha} = 2\operatorname{cosec} 2$

(B) $\sin^{-1} \left(\frac{2\alpha}{1+\alpha^2} \right) = \pi - 2$

(C) $\tan^{-1} \left(\frac{2\alpha}{\alpha^2 - 1} \right) = 2 - \pi$

(D) $\cos^{-1} \left(\frac{1-\alpha^2}{1+\alpha^2} \right) = 2$

Space for Rough Work

8. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2([x] - \{x\})$ and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = 2\left[\frac{x}{2}\right] - \frac{x}{2}$, then which of the following holds good ?
(where $[.]$ and $\{.\}$ denotes greatest integral function & fraction part function respectively)
- (A) Area of triangle formed by $y = fog(x)$, $x = 0$, $y = 0$ is 2.
(B) Area of triangle formed by $y = gof(x)$, $x = 0$, $y = 0$ is 2.
(C) $f(x)$ is bijective function.
(D) $g(x)$ is bijective function.
9. If a straight line passes through the point $P(2, 0)$ having equation $y = 2020(x - 2)$ intersects the curve $y^2 = 8x$ at the points Q, R (let $PQ = \ell_1$ & $PR = \ell_2$). Also given that G, A represents geometric and arithmetic means of ℓ_1 and ℓ_2 respectively, then correct option(s) is/are
- (A) $\ell_1, \ell_2 \geq 2$ (B) $\ell_1, \ell_2 < 2$ (C) $G \propto A$ (D) $G \propto \sqrt{A}$

Space for Rough Work

10. Let two parabolas have a common axis where focus of each being exterior to the other and let ℓ_1, ℓ_2 be their latus rectums then the locus of the mid points of the intercepts between the parabolas made on the lines parallel to the common axis is a

- (A) Straight line if $\ell_1 = \ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$ (B) Parabola if $\ell_1 \neq \ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$
(C) Parabola $\forall \ell_1, \ell_2 \in \mathbb{R}^+$ (D) Hyperbola if $\ell_1 = 2\ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$

11. If $T_n = \begin{vmatrix} n-1 & n-2 & n-6 \\ 2n-4 & 2n-6 & 2n-11 \\ 3n-9 & 3n-12 & 3n-18 \end{vmatrix}$ where $n \in \mathbb{N}$, then which of the following is/are true ?

- (A) $\prod_{n=1}^m T_n = 6^m$ (B) $\sum_{n=1}^{10} T_n = 60$ (C) $\frac{T_{n+1}}{T_n} = 2$ (D) $T_{100} \cdot T_{101} = 48$

12. Let (x, y) denote a solution of equation $\tan^{-1} x + \cot^{-1} y = \sin^{-1} \frac{3}{\sqrt{10}}$, where $x, y \in \mathbb{N}$. Value of $x + y$ can be -

- (A) 3 (B) 6 (C) 8 (D) 9

Space for Rough Work

SECTION-II : (Maximum Marks : 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. The number of solutions of the equation $\cos^{12}x + 3\cos^4x\sin^3x = \sin^9x - 1$ in $x \in [0, 100\pi]$ is
2. In a ΔABC , $\tan A = 2$, $\tan B = \frac{3}{2}$ and $c = 2\sqrt{65}$. If circumradius of the triangle is k , then value of k is
3. If a, b, c be last three digits of the number $(6! + 1)^{6!}$, then $(a + b + c)$ is

Space for Rough Work

4. Number of divisors of $1024 \times (225)^2 \times (25)^7$ which are of the form $3k + 1$ (where k is a whole number) is equal to -
5. If $f\left(x + \frac{1}{2}\right) + f\left(x - \frac{1}{2}\right) = f(x) \quad \forall x \in \mathbb{R}$, then period of function $f(x)$ is
6. If $f(x) = 2x - 1$ then number of positive solution(s) of the equation $|f(x)| = |f(|x| - 1)|$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator